RUPRECHT-KARLS-UNIVERSITÄT FAKULTÄT FÜR MATHEMATIK UND INFORMATIK HEIDELBERG

FAKULTÄT FÜR MATHEMATIK UND INFORMATIK

UNIVERSITÄT HEIDELBERG
ZUKUNFT SEIT 1386

TRANSCRIPT OF RECORDS

Last Name:
Singhal

First Name(s):
Arnav

Date of Birth:
12 November 1998

Place of Birth:
Ghazabad, Uttar Pradesh

Degree:
Master of Science (M.Sc.)

Student Number:
3767240

Degree Programme:
Data and Computer Science

Certificate Date:
24 March 2026

ID	Degree Programme/Module/Examination/Course	Semester	Grade	CP/ECTS	
1250 88 279	**Data and Computer Science M.Sc. - PO 20212**		**2.3**	**120**	
CM	**Compulsory Modules**	**Winter 25/26**		**46**	
IMS	Master's Advanced Seminar — Retrieval Augmented Generation	Summer 24	2.0	4	
IMP	Master's Advanced Practical — Master Practical "Intelligent Systems"	Winter 24/25	1.0	8	
IMT	Master's Thesis — Master's Thesis	Summer 25	2.0	30	
IMC	Master's Colloquium — Master's Colloquium	Summer 25	2.0	4	
EM	**Elective Modules**	**Summer 25**		**56**	
VC	**Visual Computing**	**Summer 24**		**22**	
IGMA	Geometric Modeling and Animation — Lecture: Geometric Modeling and Animation	Winter 23/24	3.7	8	
ISV	Scientific Visualization — Lecture: Scientific Visualization	Winter 23/24	3.3	8	
VC1	Visual Computing 1 — Computer Vision (MVSpec)	Summer 24	2.3	6	
SE	**Software Systems and Engineering**	**Winter 24/25**		**6**	
IPM	IT Project Management — IT Project Management	Winter 23/24	3.3	3	
ISWKM	Knowledge Management and Decisions in Software Engineering — Knowledge Management and Decision-Making in Software Engineering	Winter 24/25	3.7	3	
AM	**Algorithmic Data Analysis and Machine Learning**	**Summer 25**		**14**	
IML	Machine Learning				*)
	— Machine Learning, Data Mining and Predictive Modeling	Summer 25	1.3	8	
	— Exercise for Machine Learning Essentials	Summer 24	BE		
INLPT	Natural Language Processing with Transformers — Natural Language Processing with Transformers	Winter 23/24	3.7	6	

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ID	Degree Programme/Module/Examination/Course	Semester	Grade	CP/ECTS
BM	**Bachelor Modules**	**Summer 24**		**8**
ICG	Computer Graphics			
	— Computer Graphics	Summer 24	4.0	8
	— Exercise for Computer Graphics	Summer 24	BE	0
IC	Interdisciplinary Competencies			
	— Language Course International Study Centre of Heidelberg University; General German Language Course, CEFR Level A1.1, Grade 2.3	Summer 24	BE	6
IAF	**Application Field**	**Winter 25/26**	**2.3**	**18**
IAF-Gen	Approved Application Field			
	— Corporate Finance	Summer 24	2.3	6
	— Topics in Development Economics	Summer 24	3.3	8
	— Course in Approved Application Field: Master Practical: Architectural Analysis of Vision Transformers for Helmholtz Equation Reconstruction	Winter 25/26	1.3	8
	Average Module Grades	**Winter 25/26**	**2.6**	
	Average Final Module Grades	**Winter 25/26**	**2.0**	

ID	Title of the Master's Thesis	Semester	Grade	CP/ECTS
	"Evaluating Fourier Layers in DeepPET Architecture. A Comparative Study for Forward Modeling of the 2D Helmholtz Equation"	Summer 25	2.0	30

Total Credit Points/ECTS	**120**	
Overall Grade	**2.3 (good)**	

Heidelberg, 1 April 2026

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Seal

Anke Sopka
Examination Office

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Explanations

1. The grades for module examinations are assigned according to the following scale:

   grade 1 "excellent"
   grade 2 "good"
   grade 3 "satisfactory"
   grade 4 "sufficient"
   grade 5 "fail"

   For the purpose of a more differentiated assessment, the above grades may be raised or lowered by 0.3.
   The grade 0.7 and intermediate values above 4.0 are excluded.

2. The module grade is assigned according to the following scale:

   1.0 to 1.5 "excellent"
   1.6 to 2.5 "good"
   2.6 to 3.5 "satisfactory"
   3.6 to 4.0 "sufficient"
   5.0 "fail"

   If completion of a module requires more than one examination component, the grade for the module represents the weighted average of the individual examination components. The first decimal place following the decimal separator will be taken into account without rounding. All subsequent decimal places are insignificant.

3. Study and Examination regulations of the relevant degree programme contain detailed information regarding the weighting of module grades and the calculation of the overall grade.

4. The following additional abbreviations and terms were used in this document:
   PO = Examination regulations version
   BE: pass , E: successfully completed, T = attended
   CP/ECTS: a unit of measure within the European Credit Transfer System (ECTS) representing student workload. A credit is equal to 30 hours of work.

5. The certificate date is identical to the date of completion of the last exam.

6. *) = accredited
   **) = contains accredited exams

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